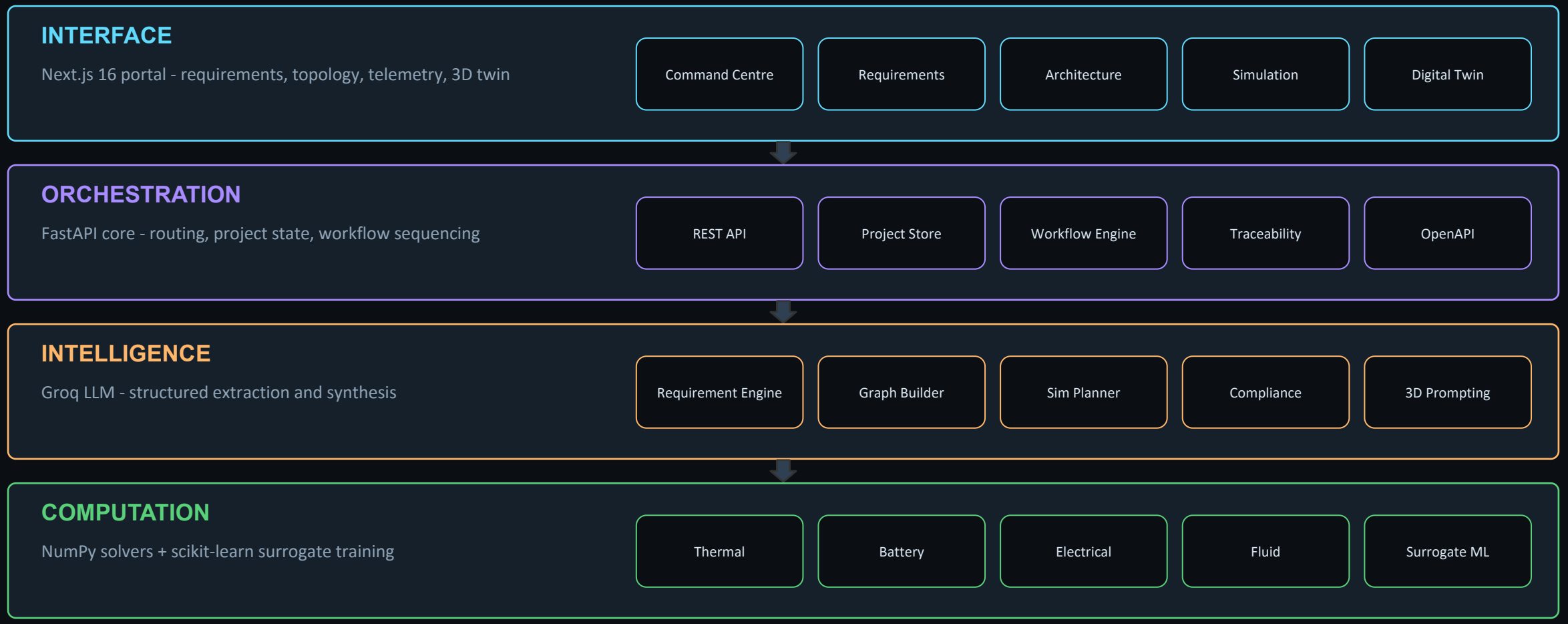
System Architecture

Anukriti AI - Generative System Engineering & Digital Twin Platform



Every LLM stage carries a deterministic fallback, so the platform runs without an API key.

Data Flow Pipeline

From one sentence of natural language to a deployable digital twin

01 **Device Description**
Natural language input from the engineer
— User input

02 **Requirement Engine**
Typed requirements, IEC safety classes, numeric constraints
— Groq LLM

03 **Graph Builder**
Subsystems, interfaces and traceability edges as a DAG
— Groq + NetworkX

04 **Simulation Manager**
Selects physics models and derives their parameters
— Groq LLM

05 **Numerical Engine**
Thermal, battery, electrical, fluid, leakage solvers
— NumPy

06 **Training Pipeline**
GradientBoosting + MLP surrogate ensemble
— scikit-learn

07 **Digital Twin**
Serialised .pkl artefact served for live inference
— joblib + FastAPI

Deployment Topology

The complete platform runs on a single Raspberry Pi 5

CLIENT

- Browser on any LAN device
- Portal UI - port 3000
- Three.js 3D rendering
- No install required

EDGE DEVICE

- Raspberry Pi 5 - 8 GB
- Next.js server - :3000
- FastAPI core - :8003
- NumPy + scikit-learn
- Models persisted as .pkl

CLOUD

- Groq inference API
- openai/gpt-oss-20b
- Structured JSON only
- No patient data leaves device

8 GB
RAM on device

4
ARM Cortex-A76 cores

64-bit
Raspberry Pi OS (Debian 13)

0x0
throttle flags - no power faults

Verified on a Raspberry Pi 5 (8 GB) with the official active cooler: the full stack installs and serves from the device.